Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/01/2021
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/31/2022 per NSSC information. (Ed., 1/31/22) NOTE: End date changed to 02/28/2022 per NSSC information. (Ed., 12/1/21) NOTE: End date changed to 8/31/2021 per D. Stocker/GRC and NSSC information (Ed., 4/20/2020) NOTE: Period of performance changed to 3/1/2017-2/28/2020 (original POP was 1/1/2017-12/31/2019) per D. Stocker/GRC (Ed., 1/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX17AD49A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Coflow Laminar Diffusion Flame" grant NNX11AP43A, with investigators Drs. Marshall Long and Mitchell Smooke. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	This past year significant advancements were made in thin filament pyrometry (TFP) processing and analysis. The TFP experiments provide the foundation for quantitative flame measurements because they allow an absolute light calibration of the Advanced Combustion via Microgravity Experiments (ACME) camera and optics. This is needed for calculating soot volume fractions, CH* mole fractions, and for improved signal-to-noise temperature measurements, by providing reference parameters for intensity-ratio pyrometry. The improvements to the TFP analysis are discussed below and have led to higher confidence in the calibration of quantitative microgravity measurements. New Demosaicing Method for TFP Analysis The color data camera used in the ACME experiments uses a Bayer color filter array, consisting of a repeated pattern of two-by-two matrices of red, green, green, and blue color filters. Since individual camera pixels only detect a single color, an interpolation must be done to infer all three color components at each pixel location – a process known as demosaicing. A common issue arises when imaging small objects whose image is near to or smaller than the pixel spacing. The SiC fiber used in the ACME experiments are 14 μm in diameter and, depending on the zoom and focus settings, the fiber image on the chip may be on the same order as the camera's 6.5 μm pixel spacing. Because the pixels associated with the red and blue color channels are only present in every other row, the image of the fiber can be absent from these channels. Standard demosaicing algorithms (e.g., the bilinear method) interpolate in both the horizontal and vertical directions to fill in the missing pixels. Near the edge of the fiber image, this leads to an interpolation of both high and low signals and creates undesirable checkered patterns, which results in unreliable fiber temperature calculations. Different methods have been used in the past to avoid these issues and to more accurately capture the fiber signal in all three color channels. The first method is to slightly defocus the camera so that the fiber image spreads across more rows of the camera sensor. This is a simple method and has proven to be useful for accurately calculating fiber temperatures. However, during some of the microgravity tests, the fiber was too well focused to provide accurate fiber temperatures. Another method was to avoid demosaicing altogether and to generate half-resolution color images with no interpolation. This method produces improved temperature measurements but some form of predicting the missing signal is still needed when performing a light calibration. To resolve these issues, a new two-step linear demoscaicing method was developed. It is similar to the bilinear method but biases the demosaicing primarily in one direction. Choosing the direction along the length of the fiber avoids the checkered patterns that were created when using the bilinear method and still produces full resolution images. While the issue of unreliable temperature results when the fiber images are smaller than or equal to the pixel spacing is not eliminated, the method does expand the limits of how focused the fiber image can be to give quality fully demosaiced images that provide meaningful temperatures. Improvements to the Fiber Light Calibration Analysis Ground-based measurements on the Yale coflow burner were used to benchmark the light calibration analysis used in the Coflow Laminar Diffusion (CLD) Flame microgravity experiments. The Yale coflow burner is known to produce stable flames and has a long history of being used in published combustion studies at Yale and other labs worldwide. Stable flame conditions that have already been studied were used so that results using the latest methods could be compared with those previously published. The tests involved varying the camera focus and fiber sweeping speed. It was found that even the slightest bit of defocusing made significant improvements to the final results. When the fiber image was spread over one to three pixels, the light calibration results were poor and usually unreliable. However, fiber images spread over five to ten pixels were greatly improved. Increasing the level of defocusing further was found to have negative results. In these overly defocused cases, the fiber temperature vs green signal began to diverge from the theoretical values based on Planck's Law. This especially becomes a problem when performing an intensity ratio pyrometry calculation since the calibration isn't able to represent the data well through the entire temperature profile. Decreasing the speed at which the fiber swept through the flame also improved the calibration results, since this created more overlap between sequential fiber images. The idea is to make sure the distance traveled by the fiber between images

	is smaller than the pixel spacing. This gives the reconstruction processing more values to average over and aids in cleaning the regions where fiber signals are missing when the camera is focused. This was found to produce cleaner signals and the fiber data matched more closely to theoretical values formulated by Planck's Law. The absolute light calibration was evaluated by plotting the fiber color-ratio temperature against the fiber green signal and comparing those results to theoretical formulations based on Planck's law of gray body light emission. The light calibration constant was calculated from the fiber temperature and green signal data at each pixel using two methods: one based on taking the mean and another using a non-linear least squares formulation. While the least squares method did the best at fitting the data, it was determined that taking the mean actually did a better job of calculating the true calibration constant with fiber tests that used more focused images. For focused fiber images, the least squares method biased towards unreliable regions of the fiber image where missing signals were present. Therefore, it was decided that using the mean produced the most reliable results overall. Preparation for CLD Flame Round Three After it was announced that there would be a third round of CLD Flame testing, new test matrices were created. The new test matrices contain two days of highly diluted ethylene flame tests and two days of highly diluted methane flame tests. In both cases, the objective is to utilize more stable mass flow controllers so that the dilution limits of both fuels in microgravity can be obtained. This was not possible during the first two rounds due to unstable behavior of some of the mass flow controllers. Further, round three will allow filling in missing test points for weak flames that were unsuccessful due to the mass flow controller issues. The test matrix also contains fiber and flame tests using a 430 nm filter. Using this filter will improve the measurements of CH*, which emits light at a wavelength of 431 nm. The ACME camera's blue channel naturally passes the majority of the light around 431 nm but transmits a broader spectrum than the 430 nm filter. This will be especially useful for the ethylene flames, which have a greener hue than the methane flames, caused by chemiluminescence from CO2*. Using the 430 nm filter will isolate the chemiluminescence from CH* and filter out other unwanted emissions. Testing with the ground copy of the ACME camera fitted with the 430 nm filter was done to determine what parameters should be used for the flight tests.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Tinajero, J, Giassi, D, Stocker, D, Long M. "Evaluation of Thin Filament Pyrometry for Light Calibration of Microgravity Coflow Flame Experiments." Pre-recorded video presentation, 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Articles in Peer-reviewed Journals	Kempema NJ, Dobbins RR, Long MB, Smooke MD. "Constrained-temperature solutions of coflow laminar diffusion flames." Proceedings of the Combustion Institute. 2021 Feb;38(2):1905-12. https://doi.org/10.1016/j.proci.2020.06.034 , Feb-2021